MATHEMATICS II FOR DATA SCIENCE
GA-1 SOLUTIONS MAY 2024

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1. Question

Match the systems of linear equations in Column A with their number of solutions in column B and their geometric representation in Column C and answer the following question.

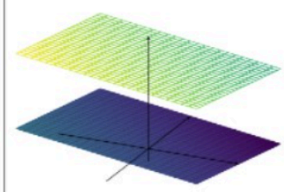
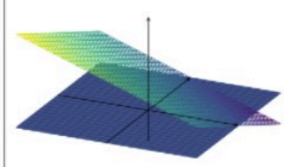
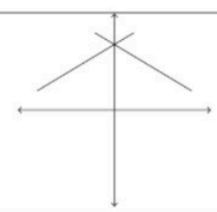
	System of linear equations (Column A)		Number of solutions (Column B)		Geometric representations (Column C)
i)	$x + y = 3,\ x - y = -3$	a)	Infinite solutions	1)	
ii)	$x + y + z = 1,\ x + y + z = 7$	b)	Unique solution	2)	
iii)	$z = 0,\ x + y + z = 1$	c)	No solution	3)	

Table: M2W1G2

Which of the following are true?

- ☐ i) $\rightarrow$ b) $\rightarrow$ 3)
- ☐ i) $\rightarrow$ a) $\rightarrow$ 3)
- ☐ ii) $\rightarrow$ c) $\rightarrow$ 2)
- ☐ ii) $\rightarrow$ c) $\rightarrow$ 1)
- ☐ iii) $\rightarrow$ a) $\rightarrow$ 2)
- ☐ iii) $\rightarrow$ a) $\rightarrow$ 1)

1.1. Answers:

- i) $\rightarrow$ b) $\rightarrow$ 3)

- ii) $\rightarrow$ c) $\rightarrow$ 1)
- iii) $\rightarrow$ a) $\rightarrow$ 2)

1.2. Solution

i)

- $\mathbf{x+y=3}$ and $\mathbf{x-y=-3}$ are the equations of straight lines.
- From $x+y=3$, we can see that $x=3-y$
- Substitute $\mathbf{x=3-y}$ in $\mathbf{x-y=-3}$

$$\begin{aligned} 3-y-y &= -3 \\ -2y &= -6 \\ y &= 3 \end{aligned}$$

- Now,

$$\begin{aligned} x &= 3-y \\ x &= 3-3 \\ x &= 0 \end{aligned}$$

- This shows that both the equations intersect at $(0,3)$. There is no other solution for these two system of linear equations. Hence, they have a unique solution.
- The straight lines formed by these two linear equations intersect at $Y-axis$, at $(0,3)$.

ii)

- $x+y+z=1$ and $x+y+z=7$ are planes in a three dimensional space.
- If for some point (x,y,z), $x+y+z$ is 1, then it is not possible for the same point to have $x+y+z$ as 7 and vice-versa. Hence, there is no solution for these two system of linear equations.
- When we look at the coefficients of both the equations (consider $a, b\ and\ c$ to be the coefficients of an equation of the form $ax+by+cz=d$),
  $\frac{a1}{a2}=\frac{b1}{b2}=\frac{c1}{c2},\ i.e.,\ \frac{1}{1}=\frac{1}{1}=\frac{1}{1}\ and\ d1 \neq d2,\ i.e.,\ 1 \neq 7$. So, these two equations would form two planes parallel to each other.

iii)

- $z=0$ is the equation of the $XY\,plane$ in three dimensional space. The intersection of $x+y+z=1$ and the $XY\,plane$ is a straight line. (With $z=0$, $x+y+z=1$ will become $x+y=1$)
- For every point in this line, the Z co-ordinate would be zero. The intersection of these two planes is a straight line. As a result, these two system of linear equations have infinite solutions.

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2. Question

Suppose $det(4A) = n \times det(A)$ for any 5×5 real matrix A. What is the value of n?

2.1. Answer : 1024

2.2. Solution

We know that the scalar multiplication of a row by a constant t multiplies the determinant by t. On multiplying 4 to matrix A, we can see that every row of matrix A is multiplied by 4. A is a 5×5 matrix. So there are 5 rows. Hence, the determinant of A gets multiplied by 4^5, which is 1024.

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3. Question

Let A be a square matrix such that $A^2 = A$. If $(I + A)^3 - 10A = mA$, then find the value of m.

3.1. Answer : $m = -3$

3.2. Solution

Given, $A^2 = A$

Properties to remember,

- Multiplication of any real matrix M with the identity matrix will result in M, provided the dimensions of the two matrices are compatible with each other.
- Multiplication of the identity matrix with any real matrix M will result in M, provided the dimensions of the two matrices are compatible with each other.

$$\begin{aligned}
(I + A)^3 - 10A &= I + mA \\
\left(I + A + A + A^2\right)(I + A) - 10A &= I + mA \\
\left(I + A + 3A + 3A^2\right) - 10A &= I + mA \\
(I + 7A) - 10A &= I + mA \\
-3A &= mA \\
\\
\implies m &= -3
\end{aligned}$$

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4. Question

If $A = \begin{bmatrix} 20 & 30 & 40 \\ 8 & 16 & 24 \\ 8 & 10 & 12 \end{bmatrix}$, then what will be the determinant of A?

4.1. Answer : 0

4.2. Solution

$det(A) = 20((16 \times 12) - (24 \times 10)) - 30((8 \times 12) - (24 \times 8)) + 40((8 \times 10) - (16 \times 8))$
$det(A) = 20(192 - 240) - 30(96 - 192) + 40(80 - 128)$
$det(A) = 20(-48) - 30(-96) + 40(-48)$
$det(A) = -960 + 2880 - 1920$
$det(A) = 0$

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5. Question

Let A be a square matrix of order 3 and B be a matrix that is obtained by adding 6 times the first row of A to the third row of A and adding 6 times the second row of A to the first row of A. If $det(A) = 5$, then find out the value of $det\left(4A^2B^{-1}\right)$

5.1. Answer : 320

5.2. Solution

Some properties to remember:

(1) Switching two rows or columns changes the sign.
(2) Multiples of rows and columns can be added together without changing the determinant's value.
(3) Scalar multiplication of a row by a constant t multiplies the determinant by t.
(4) A determinant with a row or column of zeros has value 0.
(5) $det(A)*det\left(A^{-1}\right) = det(I),\ for\ any\ matrix\ A\ with\ det(A) \neq 0$
(6) $det(kA) = k^n det(A),\ where\ n\ is\ the\ order\ of\ the\ square\ matrix\ A$

From property (2), adding 6 times of row 1 to row 3 and adding 6 times of row 2 to row 1 would not change the determinant's value. Hence, the determinant of B is also 5 and $det\left(B^{-1}\right) = \frac{1}{5}$.

$$\begin{aligned} det\left(4A^2B^{-1}\right) &= det(4A)det(A)det\left(B^{-1}\right) \\ &= 4^3 det(A) \times 5 \times \frac{1}{5} \\ &= 64 \times 5 \\ &= 320 \end{aligned}$$

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6. Question

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, then what will be the value of the sum of the diagonal elements of A^7?

6.1. Answer : 2187

6.2. Solution

$$A^1 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix},\ A^2 = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix},\ A^3 = \begin{bmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \\ 9 & 9 & 9 \end{bmatrix},\ A^4 = \begin{bmatrix} 27 & 27 & 27 \\ 27 & 27 & 27 \\ 27 & 27 & 27 \end{bmatrix},\ A^5 = \begin{bmatrix} 81 & 81 & 81 \\ 81 & 81 & 81 \\ 81 & 81 & 81 \end{bmatrix}$$

$$A^6 = \begin{bmatrix} 243 & 243 & 243 \\ 243 & 243 & 243 \\ 243 & 243 & 243 \end{bmatrix},\ A^7 = \begin{bmatrix} 729 & 729 & 729 \\ 729 & 729 & 729 \\ 729 & 729 & 729 \end{bmatrix}$$

It is easy to see that for the given matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, all the elements in A^k would be 3^{k-1}, where $k \geqslant 1\ and\ k \in \mathbb{Z}$. Sum of the diagonal elements of A^7 is $3 \times 729 = 2187$.

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7. Question

Let $A = [\alpha_{ij}]$ be a square matrix of order 3, where $a_{_{ij}} = 1i + 3j$. Find $det(A)$.

7.1. Answer : 0

7.2. Solution

Let us first find how the matrix A looks like. Given, $a_{_{ij}} = 1i + 3j$.

$$A = \begin{bmatrix} 1(1)+3(1) & 1(1)+3(2) & 1(1)+3(3) \\ 1(2)+3(1) & 1(2)+3(2) & 1(2)+3(3) \\ 1(3)+3(1) & 1(3)+3(2) & 1(3)+3(3) \end{bmatrix} \Longrightarrow A = \begin{bmatrix} 4 & 7 & 10 \\ 5 & 8 & 11 \\ 6 & 9 & 12 \end{bmatrix}$$

$$\begin{aligned} det(A) &= 4((8 \times 12) - (11 \times 9)) - 5((7 \times 12) - (10 \times 9)) + 6((7 \times 11) - (8 \times 10)) \\ det(A) &= 4(96 - 99) - 5(84 - 90) + 6(77 - 80) \\ det(A) &= 4(-3) - 5(-6) + 6(-3) \\ det(A) &= 0 \end{aligned}$$

Comprehension $(8, 9, 10)$

The scores of a student in mathematics, physics and chemistry in her class-12 board exams are m, p and c respectively, where each score is out of 100. She has applied for three engineering streams in a college. Each stream assigns different weights to these three subjects to calculate her final score, which is again out of 100.

For example, the weight given to mathematics, physics and chemistry could be $0.2, 0.7$ and 0.1

respectively by a stream. These weights are then multiplied with the corresponding scores to get the final score. Concretely, if the student has scored $85, 78, 40$ in the three subjects, her final score for this stream is:
$0.2 \times 85 + 0.7 \times 78 + 0.1 \times 40 = 75.6$

This is called a weighted average. Note that the weights always sum to 1. Now, the weights assigned by the three streams for mathematics, physics and chemistry, in this order, are given below:

Stream-1: $0.2, 0.7, 0.1$
Stream-2: $0.5, 0.3, 0.2$
Stream-3: $0.1, 0.4, 0.5$

The final score of the student in stream-1 is 81. It is 83 in stream-2 and 76 in stream-3. We wish to find the student's marks in the three subjects.

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8. Question

This is framed as a system of linear equations. Select all true options concerning the coefficient matrix if the vector of unknowns is given as $\begin{bmatrix} m \\ p \\ c \end{bmatrix}$. Assume that the first equation corresponds to stream-1, second to stream-2 and last to stream-3.

- ☐ The first row is $0.2, 0.7, 0.1$
- ☐ The last row is $0.1, 0.4, 0.5$
- ☐ The first column is $0.2, 0.7, 0.1$
- ☐ The middle column is $0.5, 0.3, 0.2$

8.1. Answers :

- The first row is $0.2,\ 0.7,\ 0.1$
- The last row is $0.1,\ 0.4,\ 0.5$

8.2. Solution

- Equation for Stream-1 $0.2m + 0.7p + 0.1c = 81 \qquad (1)$
- Equation for Stream-2 $0.5m + 0.3p + 0.2c = 83 \qquad (2)$
- Equation for Stream-3 $0.1m + 0.4p + 0.5c = 76 \qquad (3)$

Matrix form $(Ax = b)$ for the system of linear equations given above:

$$A = \begin{bmatrix} 0.2 & 0.7 & 0.1 \\ 0.5 & 0.3 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}, \quad x = \begin{bmatrix} m \\ p \\ c \end{bmatrix}, \quad b = \begin{bmatrix} 81 \\ 83 \\ 76 \end{bmatrix}$$

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9. Question

Find $\dfrac{m+p+c}{3}$.

9.1. Answer : 80

9.2. Solution

- From (1),

$$\begin{aligned} 0.1c &= 81 - 0.7p - 0.2m \\ c &= 810 - 7p - 2m \end{aligned}$$

- Substitute c in (2),

$$\begin{aligned} 0.5m + 0.3p + 0.2(810 - 7p - 2m) &= 83 \\ 0.5m + 0.3p + 162 - 1.4p - 0.4m &= 83 \\ 0.1m &= 1.1p - 79 \\ m &= 11p - 790 \end{aligned}$$

- Substitute c in (3),

$$\begin{aligned} 0.1m + 0.4p + 0.5(810 - 7p - 2m) &= 76 \\ 0.1m + 0.4p + 405 - 3.5p - m &= 76 \\ -3.1p - 0.9m &= -329 \end{aligned}$$

- Substitute $\mathbf{m = 11p - 790}$ in $\mathbf{-3.1p - 0.9m = -329}$,

$$\begin{aligned} -3.1p - 0.9(11p - 790)) &= 76 \\ -3.1p - 9.9p + 711 &= -329 \\ -13p &= -1040 \\ p &= 80 \end{aligned}$$

- Substitute $\mathbf{p = 80}$ in $\mathbf{m = 11p - 790}$,

$$
\begin{aligned}
m &= 11(80) - 790 \\
m &= 880 - 790 \\
m &= 90
\end{aligned}
$$

- Substitute $\mathbf{p = 80}$ and $\mathbf{m = 90}$ in $\mathbf{c = 810 - 7p - 2m}$,

$$
\begin{aligned}
c &= 810 - 7(80) - 2(90) \\
c &= 810 - 560 - 180 \\
c &= 70
\end{aligned}
$$

$$
\boxed{\frac{m+p+c}{3} = \frac{90+80+70}{3} = \frac{240}{3} = 80}
$$

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10. Question

Find $det(A)$, where A is the coefficient matrix.

10.1. Answer : -0.13

10.2. Solution

$$
\begin{aligned}
det(A) &= 0.2((0.3 \times 0.5) - (0.4 \times 0.2)) - 0.5((0.7 \times 0.5) - (0.4 \times 0.1)) \\
&\quad +0.1((0.7 \times 0.2) - (0.3 \times 0.1)) \\
det(A) &= 0.2(0.15 - 0.08) - 0.5(0.35 - 0.04) + 0.1(0.14 - 0.03) \\
det(A) &= 0.2(0.07) - 0.5(0.31) + 0.1(0.11) \\
det(A) &= 0.014 - 0.155 + 0.011 \\
det(A) &= 0.025 - 0.155 \\
det(A) &= -0.13
\end{aligned}
$$